Project Title - Seasonal Agriculture Performance Analysis

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PROBLEM STATEMENT

- Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions. As a result, agricultural performance may differ from one season to another. However, raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed in different seasonal conditions. The problem is to analyze the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data.

Project Description

This project focuses on analyzing agricultural data across different seasons, geographical areas, and farming conditions. The analysis aims to identify seasonal patterns, trends, relationships, and variations in agricultural performance.

The project involves data exploration, cleaning, statistical analysis, and visualization to understand how factors such as environmental conditions, farming practices, resource usage, crop production, and economic performance vary across seasons.

The findings will help generate data-driven insights and recommendations for better understanding and planning of seasonal agricultural activities.

WHO ARE THE END USERS?

- **Farmers** – To understand seasonal crop performance and improve farming decisions.
- **Agricultural Planners** – To support seasonal planning and resource allocation.
- **Government & Agricultural Departments** – To understand agricultural trends and support planning.
- **Agricultural Analysts/Researchers** – To analyze seasonal patterns and agricultural performance.
- **Agribusiness Stakeholders** – To understand production and economic performance across seasons.

Technology Used

- **Python** – Data cleaning, analysis, and processing
- **Pandas** – Data manipulation and exploration
- **NumPy** – Numerical computations
- **Matplotlib / Seaborn** – Data visualization
- **Jupyter Notebook** – Performing and documenting the analysis

RESULTS

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
#load the dataset
df = pd.read_csv("C:/Users/bisha/OneDrive/Desktop/powerBI/seasonal_agriculture_performance_dataset.csv")
print("Dataset loaded successfully!")
```

```
Dataset loaded successfully!
```

```
print("Number of rows:", df.shape[0])
print("Number of columns:", df.shape[1])
```

```
Number of rows: 4000
Number of columns: 28
```

```
df.head()
```

	Farm_ID	State	District	Crop	Season	Farm_Area_Hectares	Rainfall_mm	Avg_Temperature_C	Humidity_pc
0	SF10001	Andhra Pradesh	Rajkot	Wheat	Kharif	0.53	486.4	24.3	69.
1	SF10002	Maharashtra	Nalgonda	Maize	Kharif	6.53	855.4	25.7	78.

```
df.tail()
```

	Farm_ID	State	District	Crop	Season	Farm_Area_Hectares	Rainfall_mm	Avg_Temperature_C
3995	SF13996	Karnataka	Rajkot	Wheat	Kharif	4.72	865.4	27.3
3996	SF13997	Karnataka	Krishna	Chilli	Kharif	8.77	947.5	29.2
3997	SF13998	Andhra Pradesh	Erode	Pulses	Zaid	10.47	80.0	35.4
3998	SF13999	Punjab	Warangal	Cotton	Kharif	1.73	475.0	28.2
3999	SF14000	Punjab	Nashik	Wheat	Kharif	3.31	984.3	25.8

5 rows × 28 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4000 entries, 0 to 3999
Data columns (total 28 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Farm_ID             4000 non-null   object
 1   State               4000 non-null   object
 2   District            4000 non-null   object
```

RESULTS

```
missing_values = df.isnull().sum()

print("Missing values in each column:")
print(missing_values)
```

```
Missing values in each column:
Farm_ID                 0
State                   0
District                0
Crop                    0
Season                  0
Farm_Area_Hectares      0
Rainfall_mm            48
Avg_Temperature_C       0
Humidity_pct            0
Sunlight_Hours_Day      0
Soil_pH                 0
Soil_Moisture_pct      40
Nitrogen_kg_ha          0
Phosphorus_kg_ha        0
Potassium_kg_ha         0
Irrigation_Method       0
Fertilizer_kg_ha        0
Pesticide_Litre_ha      0
Seed_Quality_Score      0
Yield_Tonnes_Ha        32
Production_Tonnes       0
```

```
# check duplicates
duplicates = df.duplicated().sum()

print("Number of duplicate records:", duplicates)
```

```
Number of duplicate records: 0
```

```
df.describe()
```

	Farm_Area_Hectares	Rainfall_mm	Avg_Temperature_C	Humidity_pct	Sunlight_Hou
count	4000.000000	3952.000000	4000.00000	4000.000000	4000.
mean	7.888265	601.152657	26.81890	63.210500	7.
std	4.223764	288.925025	3.81818	11.960834	1.
min	0.500000	80.000000	16.20000	25.000000	3.
25%	4.260000	370.200000	23.90000	54.775000	6.
50%	7.895000	582.000000	26.90000	63.000000	7.
75%	11.650000	834.950000	29.60000	71.900000	8.
max	15.000000	1395.200000	39.70000	95.000000	11.

8 rows × 22 columns

RESULTS

```
#Unique Values
for column in df.columns:
    print("\nColumn:", column)
    print("Unique values:", df[column].nunique())
```

```
Column: Farm_ID
Unique values: 4000

Column: State
Unique values: 8

Column: District
Unique values: 10

Column: Crop
Unique values: 8

Column: Season
Unique values: 3

Column: Farm_Area_Hectares
Unique values: 1370

Column: Rainfall_mm
Unique values: 3256

Column: Avg_Temperature_C
```

```
# Numerical columns (missing value handling)
numerical_columns = df.select_dtypes(include=np.number).columns

for column in numerical_columns:
    df[column] = df[column].fillna(df[column].median())
```

```
# Categorical columns
categorical_columns = df.select_dtypes(include='object').columns

for column in categorical_columns:
    df[column] = df[column].fillna(df[column].mode()[0])
```

```
# Removes Extra Spaces from Text
for column in categorical_columns:
    df[column] = df[column].astype(str).str.strip()
```

```
print("Missing values after cleaning:")
print(df.isnull().sum())
```

```
Missing values after cleaning:
Farm_ID                   0
State                     0
District                  0
Crop                      0
Season                    0
Farm_Area_Hectares        0
Rainfall_mm               0
```

RESULTS

```
print(df['Season'].unique())
```

```
['Kharif' 'Rabi' 'Zaid']
```

```
df.columns.tolist()
```

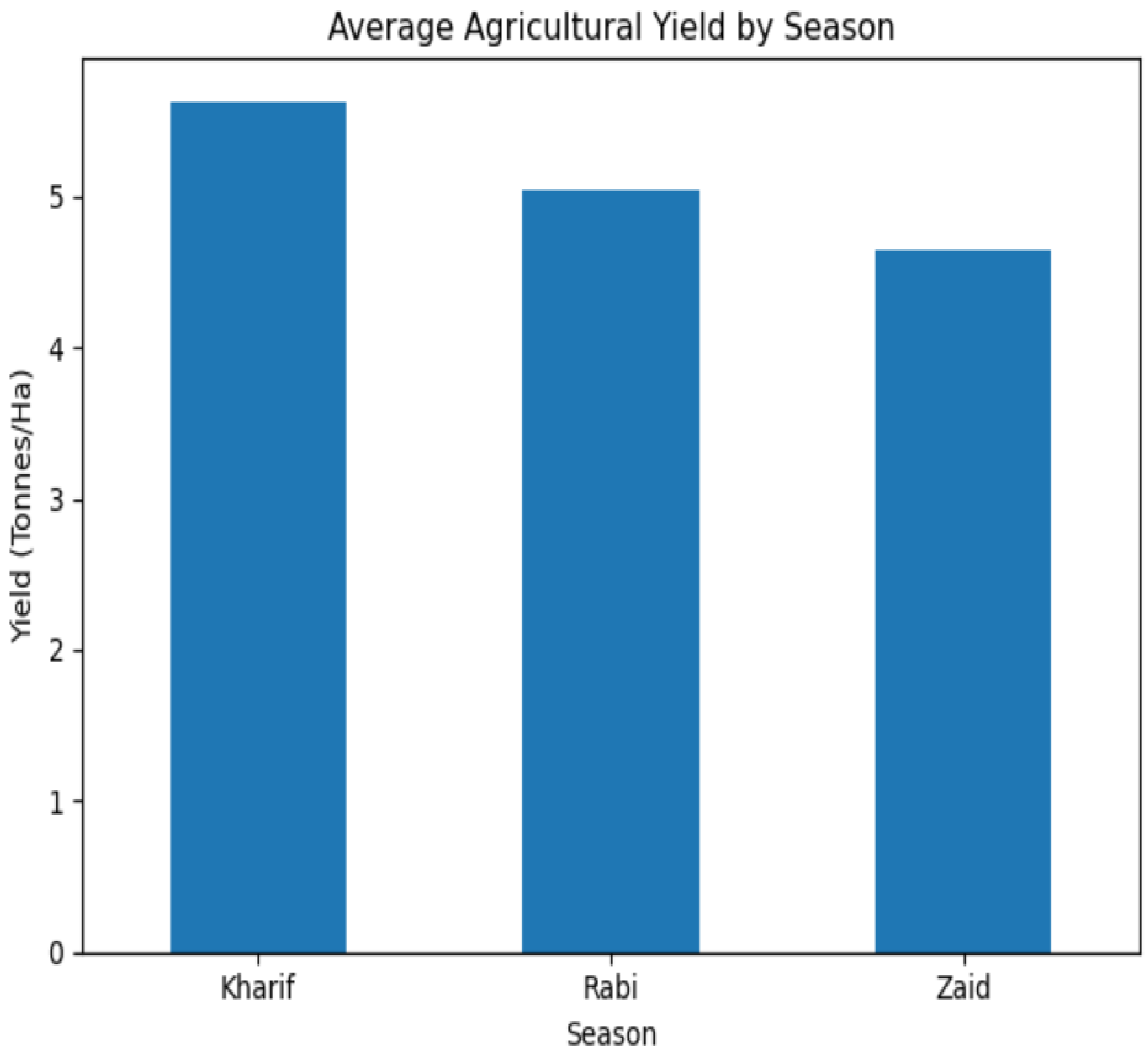
```
['Farm_ID',
 'State',
 'District',
 'Crop',
 'Season',
 'Farm_Area_Hectares',
 'Rainfall_mm',
 'Avg_Temperature_C',
 'Humidity_pct',
 'Sunlight_Hours_Day',
 'Soil_pH',
 'Soil_Moisture_pct',
 'Nitrogen_kg_ha',
 'Phosphorus_kg_ha',
 'Potassium_kg_ha',
 'Irrigation_Method',
 'Fertilizer_kg_ha',
 'Pesticide_Litre_ha',
 'Seed_Quality_Score',
 'Yield_Tonnes_Ha',
 'Production_Tonnes',
 'Market_Price_INR_Tonne',
```

```
# Compare agricultural performance across seasons

season_performance = df.groupby('Season')[[
    'Yield_Tonnes_Ha',
    'Production_Tonnes',
    'Profit_INR'
]].mean().round(2)

print(season_performance)
```

```
        Yield_Tonnes_Ha  Production_Tonnes  Profit_INR
Season
Kharif             5.63              46.31   178914.65
Rabi               5.04              41.49    87689.47
Zaid               4.64              38.89   -24804.82
```

```
season_performance['Yield_Tonnes_Ha'].plot(
    kind='bar',
    title='Average Agricultural Yield by Season',
    xlabel='Season',
    ylabel='Yield (Tonnes/Ha)'
)

plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

RESULTS




```
season_pattern = df.groupby('Season')[[
    'Yield_Tonnes_Ha',
    'Production_Tonnes',
    'Profit_INR'
]].mean().round(2)

print(season_pattern)
```

```
        Yield_Tonnes_Ha  Production_Tonnes  Profit_INR
Season
Kharif             5.63              46.31   178914.65
Rabi               5.04              41.49    87689.47
Zaid               4.64              38.89   -24804.82
```

```
season_conditions = df.groupby('Season')[[
    'Rainfall_mm',
    'Avg_Temperature_C',
    'Soil_Moisture_pct',
    'Yield_Tonnes_Ha'
]].mean().round(2)

print(season_conditions)
```

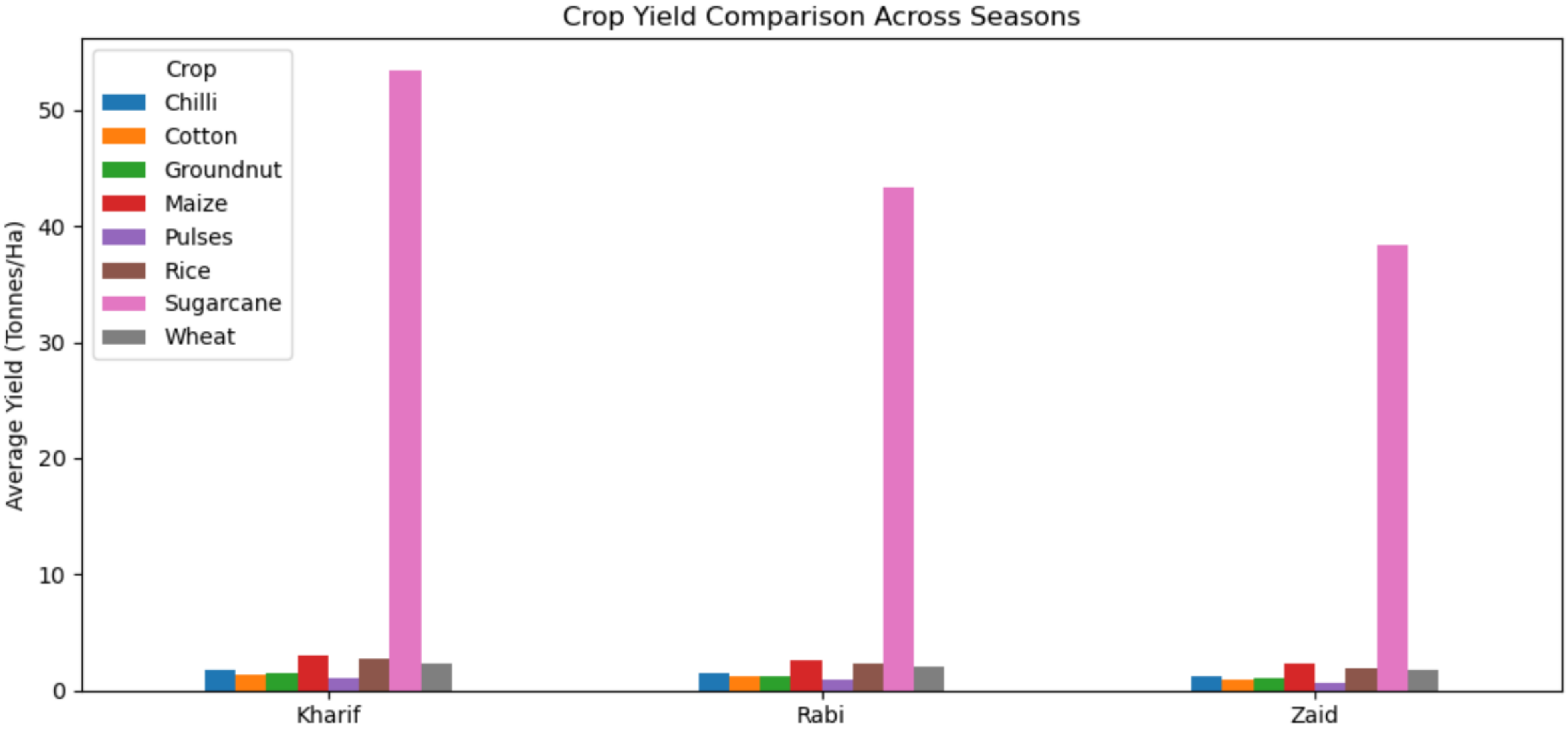
```
        Rainfall_mm  Avg_Temperature_C  Soil_Moisture_pct  Yield_Tonnes_Ha
Season
Kharif       849.20              28.45              31.15             5.63
Rabi         437.62              23.49              24.07             5.04
Zaid         304.65              31.04              19.28             4.64
```

RESULTS

```
season_crop = df.groupby(['Season', 'Crop'])['Yield_Tonnes_Ha'].mean().round(2)

print(season_crop)
```

```
Season  Crop
Kharif  Chilli        1.73
        Cotton        1.37
```

```
season_crop.unstack().plot(kind='bar', figsize=(10, 5))

plt.title('Crop Yield Comparison Across Seasons')
plt.xlabel('Season')
plt.ylabel('Average Yield (Tonnes/Ha)')
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



RESULTS

```python
# Compare yield and profit across seasons
season_analysis = df.groupby('Season')[[
    'Yield_Tonnes_Ha',
    'Profit_INR'
]].mean().round(2)

print(season_analysis)
```

```
        Yield_Tonnes_Ha  Profit_INR
Season
Kharif             5.63   178914.65
Rabi               5.04    87689.47
Zaid               4.64   -24804.82
```

```python
# Find farms with negative profit
unusual = df[df['Profit_INR'] < 0]

print(unusual[['Season', 'Crop', 'Yield_Tonnes_Ha', 'Profit_INR']].head())
```

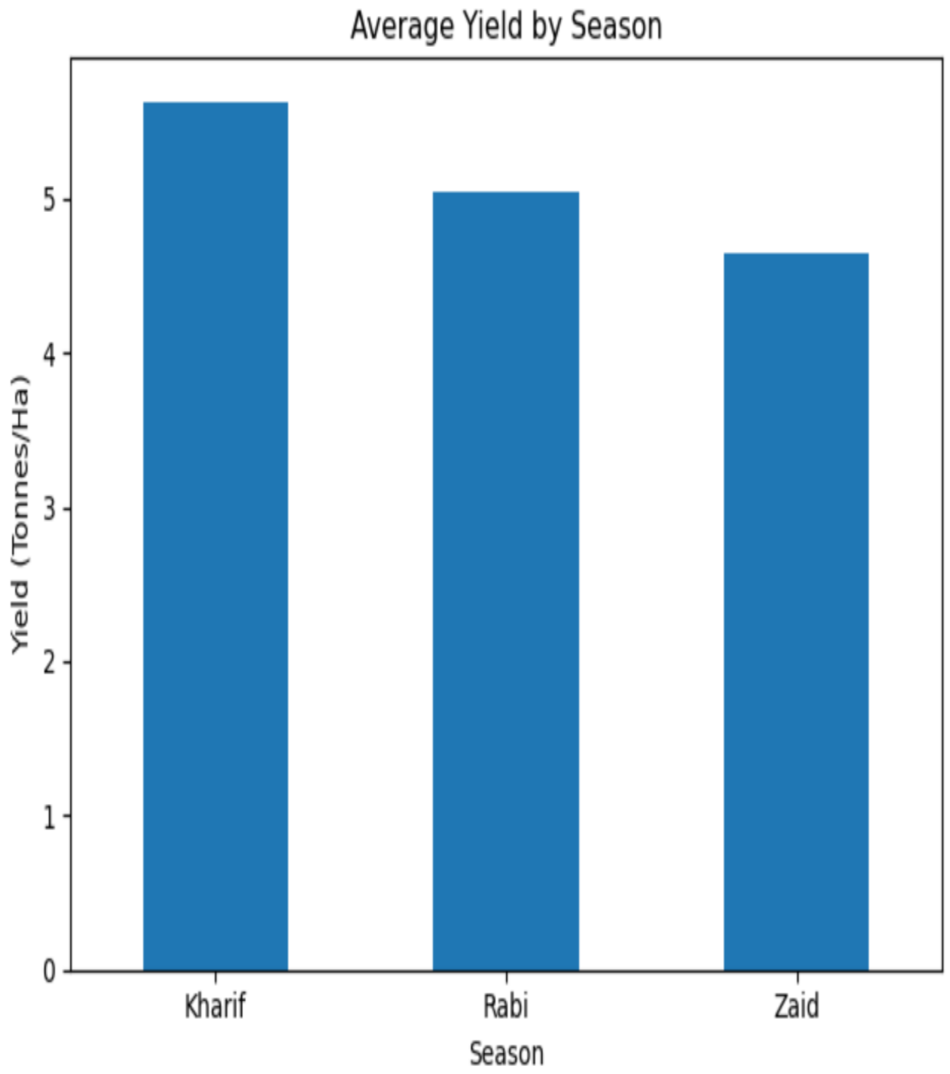
```
   Season    Crop  Yield_Tonnes_Ha  Profit_INR
0  Kharif   Wheat             2.46      -11852
1  Kharif   Maize             0.30     -451950
2    Rabi  Pulses             0.52     -124744
3  Kharif    Rice             1.84      -95704
4    Zaid   Maize             2.21     -144352
```

```python
# Statistical summary by season
stats = df.groupby('Season')[[
    'Yield_Tonnes_Ha',
    'Production_Tonnes',
    'Profit_INR'
]].agg(['mean', 'std']).round(2)

print(stats)
```

```
       Yield_Tonnes_Ha        Production_Tonnes         Profit_INR
                  mean    std              mean    std        mean        std
Season
Kharif            5.63  14.39             46.31  135.7   178914.65  611549.78
Rabi              5.04  12.75             41.49  121.9    87689.47  489901.25
Zaid              4.64  11.27             38.89  112.0   -24804.82  436356.75
```

```python
# Average yield by season
df.groupby('Season')['Yield_Tonnes_Ha'].mean().plot(
    kind='bar',
    title='Average Yield by Season',
    xlabel='Season',
    ylabel='Yield (Tonnes/Ha)'
)

plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

RESULTS




```python
# Seasonal performance summary
summary = df.groupby('Season')[[
    'Yield_Tonnes_Ha',
    'Production_Tonnes',
    'Profit_INR'
]].mean().round(2)

print("Seasonal Performance:")
print(summary)

# Find best and lowest performing seasons
best_yield = summary['Yield_Tonnes_Ha'].idxmax()
lowest_yield = summary['Yield_Tonnes_Ha'].idxmin()

best_profit = summary['Profit_INR'].idxmax()
lowest_profit = summary['Profit_INR'].idxmin()

print("\nConclusion:")
print("Highest yield season:", best_yield)
print("Lowest yield season:", lowest_yield)
print("Highest profit season:", best_profit)
print("Lowest profit season:", lowest_profit)
```

```
Seasonal Performance:
        Yield_Tonnes_Ha  Production_Tonnes  Profit_INR
Season
Kharif             5.63              46.31   178914.65
Rabi               5.04              41.49    87689.47
Zaid               4.64              38.89   -24804.82

Conclusion:
Highest yield season: Kharif
Lowest yield season: Zaid
Highest profit season: Kharif
Lowest profit season: Zaid
```

Future scope

FUTURE SCOPE

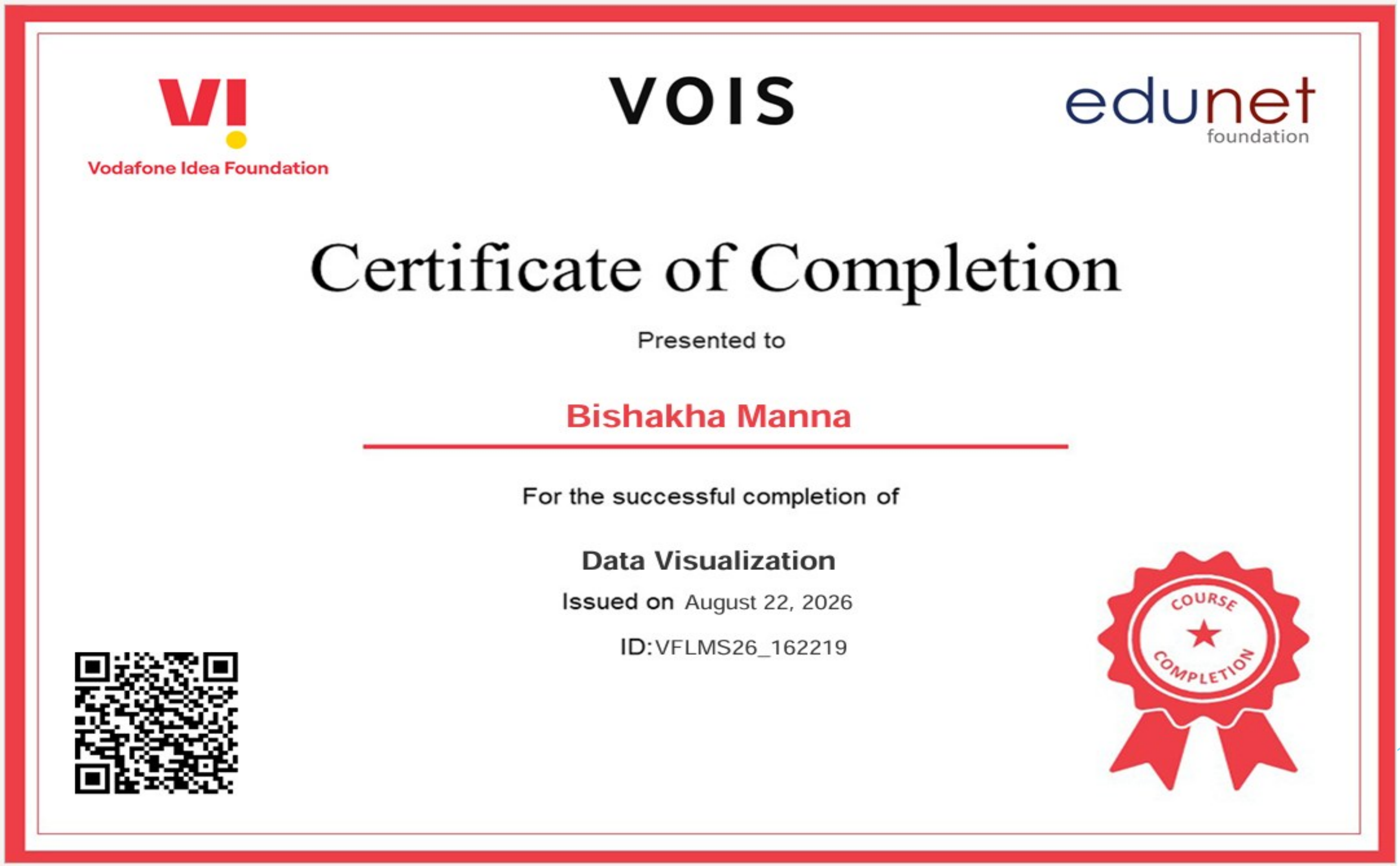
- Develop **predictive models** for seasonal crop performance.
- Include **real-time weather and market data**.
- Build an **interactive dashboard** for farmers and planners.
- Provide **personalized crop and resource recommendations**.
- Expand the analysis to **more regions and crops**.

GitHub Link

GitHub Repository Link-

https://github.com/bishakhamanna2-pixel/AgriculturePerformanceAnalysis.git

VOIS Course completion certificate - Course name- Data Visualization



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COURSE COMPLETION

Thank you